

The order of $R_e(t)$ over the sublattice subring is still infinite and the two-parameter commutator $[R_e(t), R_f(s)]$

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Abstract

Let $R_e(t)$ add a connected e -ribbon with weight t^{ht} to a Schur function, and let $\Lambda^{(e)} = \mathbb{Q}(t)[p_e, p_{2e}, \dots]$. Question Q96 asked whether the infinite differential order of $R_e(t)$ — proved over the full base Λ — is an artifact of measuring against *all* the power sums, and collapses to a finite order when the base is shrunk to the sublattice subring $\Lambda^{(e)}$.

The answer is no, and emphatically so. We prove the sharper statement that the order is already infinite over the *single-generator* subring $\mathbb{Q}(t)[p_e]$: for every $e \geq 2$, every $d \geq 1$ and $\mu_d = ((d+1)e - 1, 1)$,

$$\langle s_{\mu_d}, \text{ad}(M_{p_e})^d(R_e(t)) s_{\emptyset} \rangle = (-1)^d (1+t).$$

Since $\mathbb{Q}(t)[p_e] \subseteq \Lambda^{(e)} \subseteq \Lambda$ and order is monotone *decreasing* in the base, this forces $\text{ord}_{\mathbb{Q}(t)[p_e]} = \text{ord}_{(e)} = \text{ord} = \infty$ for every $t \neq -1$. The deformation is genuinely transverse to the sublattice; the specialness of the anchor $t = -1$ is not a choice of coordinates.

Three by-products. (i) The witness is a *single-channel* matrix element: for this source and target there is exactly one legal sequence of e -moves, forced by a flow-conservation argument on the abacus, and it is unique *as an ordered sequence*. The whole computation is therefore a $(d+1)$ -term alternating sum with two distinct weights, t and -1 . (ii) The same witness re-proves the earlier full-base result ([2, Cor. 4.9]) uniformly in the parameter, including the scalar $c = 0$, which previously needed a separate theorem. (iii) We derive the two-parameter cross-rank commutator $[R_e(t), R_f(s)]$ in closed form. Its two-bead sector depends on the two parameters only through the product ts , and vanishes identically exactly on the locus $ts = 1$; its one-bead sector is a genuinely *two-coloured* dressing — each half of the interval carries the parameter of the operator that traversed it — which is why it does not collapse to a scalar multiple of a single dressed ribbon move when $s \neq t$. We also correct a case split: $[R_e(t), R_e(s)]$ has a *nonzero* two-bead part when $t \neq s$, so the “ $e = f \Rightarrow 0$ ” clause of [3, Thm. 2.3(2)] is a one-parameter statement only.

The argument is elementary abacus combinatorics throughout. In particular it does not pass through Lam’s LLT map Φ at any point (see Remark 15), and no claim about the literature is made anywhere in this note.

Verified by two code-disjoint engines; see §5.

1 Setting

Convention 1. $\Lambda = \mathbb{Q}(t)[p_1, p_2, \dots]$ is the ring of symmetric functions over $\mathbb{Q}(t)$, with Schur basis $\{s_\lambda\}$. For a partition λ the *Maya set* is $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$; its elements are *beads*. We write $|M\rangle$ for the corresponding s_λ and identify Λ with the charge-0 semi-infinite wedge $\mathcal{F}$. For $g \geq 1$,

$$R_g(t) s_\lambda = \sum_{\mu} t^{\text{ht}(\mu/\lambda)} s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected border strip of size g and $\text{ht} = (\#\text{rows}) - 1$. M_f denotes multiplication by $f \in \Lambda$.

Proposition 2 (abacus form; [3, Prop. 1.2]). *Call (b, g) a legal g -move on the Maya set M if $b \in M$ and $b + g \notin M$, and put $M \cdot (b, g) = (M \setminus \{b\}) \cup \{b + g\}$. Then*

$$R_g(t) |M\rangle = \sum_{(b,g) \text{ legal}} t^{\#(M \cap (b, b+g))} |M \cdot (b, g)\rangle,$$

a finite sum; $b \mapsto M \cdot (b, g)$ is a bijection from legal g -moves onto the μ with μ/λ a connected g -ribbon, and $\text{ht}(\mu/\lambda) = \#(M \cap (b, b + g))$. Here $(b, b + g)$ is the open interval.

Proposition 3 (Murnaghan–Nakayama; [1, Q75]). $R_g(-1) = M_{p_g}$ for every $g \geq 1$.

This is the load-bearing input and it was re-verified at the start of the session against an independent implementation of $p_g \cdot s_\lambda$ in $\mathbb{Q}[p_1, p_2, \dots]$ (§5, item 2).

2 Order over a subring, and why the generators suffice

Definition 4 (order filtration over a base). Let $B \subseteq \Lambda$ be a $\mathbb{Q}(t)$ -subalgebra containing 1. For a $\mathbb{Q}(t)$ -linear operator X on Λ set $\mathcal{D}_B^{\leq -1} = \{0\}$ and, for $d \geq 0$,

$$\mathcal{D}_B^{\leq d} = \{X : [X, M_f] \in \mathcal{D}_B^{\leq d-1} \text{ for every } f \in B\},$$

equivalently $[\dots [[X, M_{f_0}], M_{f_1}], \dots, M_{f_d}] = 0$ for all $f_0, \dots, f_d \in B$. Put $\text{ord}_B(X) = \min\{d : X \in \mathcal{D}_B^{\leq d}\}$, and ∞ if there is no such d . We write $\text{ord} = \text{ord}_\Lambda$ and $\text{ord}_{(e)} = \text{ord}_{\Lambda^{(e)}}$ for $\Lambda^{(e)} = \mathbb{Q}(t)[p_e, p_{2e}, p_{3e}, \dots]$.

Λ is free as a $\Lambda^{(e)}$ -module (a basis is given by the monomials in the p_m with $e \nmid m$), so this is the Grothendieck order filtration of X as a differential operator on Λ *over the base* B . Domain and codomain are all of Λ throughout; we are changing the base, not restricting the domain. (Restricting the domain is not an option: $R_2(t)s_\emptyset = s_{(2)} + t s_{(1,1)} = \frac{1}{2}(p_1^2 + p_2) + \frac{t}{2}(p_1^2 - p_2) \notin \Lambda^{(2)}$, so $R_e(t)$ does not preserve $\Lambda^{(e)}$.)

Lemma 5 (monotonicity in the base). *If $B \subseteq B'$ are subalgebras of Λ then $\mathcal{D}_{B'}^{\leq d} \subseteq \mathcal{D}_B^{\leq d}$ for every d , hence $\text{ord}_B(X) \leq \text{ord}_{B'}(X)$ for every X . In particular*

$$\text{ord}_{\mathbb{Q}(t)[p_e]}(X) \leq \text{ord}_{(e)}(X) \leq \text{ord}(X).$$

Proof. Induction on d . For $d = -1$ both sides are $\{0\}$. If $X \in \mathcal{D}_{B'}^{\leq d}$ and $f \in B \subseteq B'$ then $[X, M_f] \in \mathcal{D}_{B'}^{\leq d-1} \subseteq \mathcal{D}_B^{\leq d-1}$ by the inductive hypothesis; so $X \in \mathcal{D}_B^{\leq d}$. The displayed chain is the case $\mathbb{Q}(t)[p_e] \subseteq \Lambda^{(e)} \subseteq \Lambda$. $\square$

So a lower bound proved over the *smallest* base is the strongest of the three. That is what we will prove.

The next lemma is the reduction the plan asked to be established rather than imported. It is standard for a polynomial base, but it is load-bearing here, so we prove it.

Lemma 6 (generators suffice). *Let B be generated as a $\mathbb{Q}(t)$ -algebra with 1 by a subset $S \subseteq \Lambda$. Define $\mathcal{E}^{\leq -1} = \{0\}$ and $\mathcal{E}^{\leq d} = \{X : [X, M_g] \in \mathcal{E}^{\leq d-1} \forall g \in S\}$. Then $\mathcal{E}^{\leq d} = \mathcal{D}_B^{\leq d}$ for every $d \geq -1$. Consequently $\text{ord}_B(X) \leq d$ if and only if every $(d+1)$ -fold nested bracket of X against multiplication operators M_g , $g \in S$, vanishes.*

Proof. Each $\mathcal{E}^{\leq c}$ is a $\mathbb{Q}(t)$ -subspace, since a nested bracket is linear in X .

Step 1: $\mathcal{E}^{\leq c}$ is stable under multiplication by any M_h , $h \in \Lambda$, on either side. Induct on c . For $c = -1$ there is nothing to prove. Let $Y \in \mathcal{E}^{\leq c}$ and $g \in S$. Multiplication operators commute with each other, so $[M_h, M_g] = 0$ and

$$[Y M_h, M_g] = [Y, M_g] M_h + Y [M_h, M_g] = [Y, M_g] M_h.$$

By definition $[Y, M_g] \in \mathcal{E}^{\leq c-1}$, so by the inductive hypothesis $[Y, M_g] M_h \in \mathcal{E}^{\leq c-1}$; hence $Y M_h \in \mathcal{E}^{\leq c}$. The argument for $M_h Y$ is identical.

Step 2: the induction on d . $\mathcal{E}^{\leq -1} = \mathcal{D}_B^{\leq -1}$. Assume $\mathcal{E}^{\leq d-1} = \mathcal{D}_B^{\leq d-1}$. The inclusion $\mathcal{D}_B^{\leq d} \subseteq \mathcal{E}^{\leq d}$ is immediate from $S \subseteq B$. Conversely let $X \in \mathcal{E}^{\leq d}$ and put

$$T = \{f \in \Lambda : [X, M_f] \in \mathcal{E}^{\leq d-1}\}.$$

Then T is a $\mathbb{Q}(t)$ -subspace ($f \mapsto M_f$ is linear); $1 \in T$ because $M_1 = \text{id}$ and $[X, \text{id}] = 0$; and $S \subseteq T$ by hypothesis. If $f, g' \in T$ then $M_{fg'} = M_f M_{g'}$ and

$$[X, M_{fg'}] = [X, M_f] M_{g'} + M_f [X, M_{g'}],$$

and both summands lie in $\mathcal{E}^{\leq d-1}$ by Step 1; so $fg' \in T$. Hence T is a subalgebra containing 1 and S , so $T \supseteq B$. Thus $[X, M_f] \in \mathcal{E}^{\leq d-1} = \mathcal{D}_B^{\leq d-1}$ for every $f \in B$, i.e. $X \in \mathcal{D}_B^{\leq d}$. $\square$

Corollary 7. $\text{ord}_{\mathbb{Q}(t)[p_e]}(X) \leq d$ if and only if $\text{ad}^{d+1}(X) = 0$, where

$$\text{ad}(X) := [X, M_{p_e}], \quad \text{ad}^d(X) = \sum_{r=0}^d (-1)^r \binom{d}{r} M_{p_e}^r X M_{p_e}^{d-r},$$

the second equality because the M_{p_e} commute with one another. In particular, if $\text{ad}^d(X) \neq 0$ for every $d \geq 1$ then $\text{ord}_{\mathbb{Q}(t)[p_e]}(X) = \infty$, and hence by Lemma 5 also $\text{ord}_{(e)}(X) = \text{ord}(X) = \infty$.

Proof. Lemma 6 with $S = \{p_e\}$, $B = \mathbb{Q}(t)[p_e]$. $\square$

3 The witness

Fix $e \geq 2$ and $d \geq 1$. Write $M_\emptyset = \mathbb{Z}_{<0}$ for the Maya set of the empty partition, and

$$M'_d := (\mathbb{Z}_{<0} \setminus \{-2\}) \cup \{(d+1)e - 2\}.$$

Lemma 8 (the target is a hook). M'_d is the Maya set of $\mu_d := ((d+1)e - 1, 1)$, a partition of $(d+1)e$.

Proof. List the beads of M'_d in decreasing order: $(d+1)e-2$ (which is $\geq e-2 \geq 0$, so it is the largest), then $-1, -3, -4, -5, \dots$. If β_j is the j -th largest bead then $\lambda_j = \beta_j + j$. So $\lambda_1 = (d+1)e-2+1 = (d+1)e-1$, $\lambda_2 = -1+2 = 1$, and $\lambda_j = -j+j = 0$ for $j \geq 3$. Its size is $(d+1)e$, which is also the degree by which $\text{ad}^d(R_e(t))$ raises, since each of the $d+1$ factors adds e cells. $\square$

Lemma 9 (the channel is a single forced route). *Let $e \geq 2$, $d \geq 1$. There is exactly one sequence of $d+1$ legal e -moves*

$$M_\emptyset = M^{(0)} \longrightarrow M^{(1)} \longrightarrow \dots \longrightarrow M^{(d+1)} = M'_d,$$

namely: for $k = 0, 1, \dots, d$ in this order, move the bead at $-2+ke$ to $-2+(k+1)e$. The intermediate states are $M^{(k)} = (\mathbb{Z}_{<0} \setminus \{-2\}) \cup \{-2+ke\}$.

Proof. Suppose $x_1 \rightarrow x_1 + e, \dots, x_{d+1} \rightarrow x_{d+1} + e$ is such a sequence. For $v \in \mathbb{Z}$ let

$$\delta(v) = \#\{i : x_i + e = v\} - \#\{i : x_i = v\}.$$

The occupancy of v increases by 1 each time a move lands on v and decreases by 1 each time a move departs from v , so summing over the whole sequence, $\delta(v) = \mathbf{1}_{M'_d}(v) - \mathbf{1}_{M_\emptyset}(v)$, i.e.

$$\delta = \mathbf{1}_{\{(d+1)e-2\}} - \mathbf{1}_{\{-2\}}.$$

(This is pure counting; it does not use legality.)

Every move preserves the residue class mod e . Fix a class, identify it with $\mathbb{Z}$ by $c+ke \leftrightarrow k$, and let m_k be the number of moves $c+ke \rightarrow c+(k+1)e$; only finitely many m_k are nonzero, and $\delta(c+ke) = m_{k-1} - m_k$.

If $c \not\equiv -2 \pmod{e}$ then δ vanishes on the class, so $m_{k-1} = m_k$ for all k ; with finite support this forces $m_k = 0$ for all k . So no move occurs outside the class of -2 .

Take $c = -2$, indexing so that k corresponds to the site $-2+ke$; then $\delta(-2+ke) = \mathbf{1}_{k=d+1} - \mathbf{1}_{k=0}$. For $k \leq -1$ we get $m_{k-1} = m_k$, hence $m_k = 0$ for all $k \leq -1$; at $k = 0$, $m_{-1} - m_0 = -1$ gives $m_0 = 1$; for $1 \leq k \leq d$, $m_{k-1} = m_k$ gives $m_k = 1$; at $k = d+1$, $m_d - m_{d+1} = 1$ gives $m_{d+1} = 0$, and then $m_k = 0$ for all $k > d+1$. So the multiset of moves is exactly $\{-2+ke \rightarrow -2+(k+1)e : 0 \leq k \leq d\}$, which already has $d+1$ elements — the total number of moves. Hence there are no others, in any class.

It remains to fix the order. For $k \geq 1$ the site $-2+ke$ satisfies $-2+ke \geq e-2 \geq 0$, so it is *unoccupied* in $M_\emptyset$, and the only move that can occupy it is the move indexed $k-1$. Legality of move k requires $-2+ke$ to be occupied at that instant, so move k must come after move $k-1$. Therefore the order is $0, 1, \dots, d$, and the intermediate states are as stated. $\square$

Lemma 10 (the two weights). *In the notation of Lemma 9, let $h_k = \#(M^{(k)} \cap (-2+ke, -2+(k+1)e))$ be the height picked up by move k . Then, for $e \geq 2$,*

$$h_0 = 1, \quad h_k = 0 \quad (1 \leq k \leq d).$$

Proof. $k = 0$: $M^{(0)} = \mathbb{Z}_{<0}$ and the open window is $(-2, e-2) = \{-1, 0, 1, \dots, e-3\}$. Since $e \geq 2$ we have $e-2 > -1$, so -1 lies in the window; the remaining sites are ≥ 0 and therefore unoccupied. Hence $h_0 = 1$. (For $e = 1$ the window is empty and $h_0 = 0$; this is exactly where the argument needs $e \geq 2$.)

$1 \leq k \leq d$: the open window is $\{-1+ke, \dots, -3+(k+1)e\}$, and its least element satisfies $-1+ke \geq -1+e \geq 1 > 0$. The only non-negative bead of $M^{(k)} = (\mathbb{Z}_{<0} \setminus \{-2\}) \cup \{-2+ke\}$ is $-2+ke$, which is the left endpoint and so not in the open window. Hence $h_k = 0$. $\square$

Theorem 11 (main). *For every $e \geq 2$ and every $d \geq 1$, with $\mu_d = ((d+1)e - 1, 1)$,*

$$\boxed{\langle s_{\mu_d}, \text{ad}(M_{p_e})^d(R_e(t)) s_{\emptyset} \rangle = (-1)^d (1+t).}$$

The same formula holds with t replaced by any scalar c .

Proof. By Corollary 7, $\text{ad}^d(R_e(t)) = \sum_{r=0}^d (-1)^r \binom{d}{r} M_{p_e}^r R_e(t) M_{p_e}^{d-r}$. By Proposition 3, $M_{p_e} = R_e(-1)$, so every summand is a composite of $d+1$ operators each of which acts on Maya sets by legal e -moves (Proposition 2); the two possible weights of a move of height h are t^h (if performed by $R_e(t)$) and $(-1)^h$ (if performed by M_{p_e}).

Fix r and look at the summand $M_{p_e}^r R_e(t) M_{p_e}^{d-r}$. Its $\langle M'_d | \cdot | M_{\emptyset} \rangle$ entry is the sum, over all sequences of $d+1$ legal e -moves from $M_{\emptyset}$ to M'_d , of the product of the weights. By Lemma 9 there is exactly one such sequence, and it is the same one for every r ; the operators act rightmost-first, so $R_e(t)$ performs move number $j := d - r$ (indices $0, \dots, d$ as in Lemma 9) and M_{p_e} performs all the others. By Lemma 10 the weight of that unique route is

$$w_j = t^{h_j} \prod_{i \neq j} (-1)^{h_i} = \begin{cases} t^1 \cdot 1 = t, & j = 0, \\ t^0 \cdot (-1)^1 = -1, & 1 \leq j \leq d. \end{cases}$$

Therefore, re-indexing by $j = d - r$,

$$\langle M'_d | \text{ad}^d(R_e(t)) | M_{\emptyset} \rangle = \sum_{j=0}^d (-1)^{d-j} \binom{d}{j} w_j = \underbrace{\sum_{j=0}^d (-1)^{d-j} \binom{d}{j} (-1)}_{=-(1-1)^d=0 \ (d \geq 1)} + (-1)^d \binom{d}{0} (t - (-1)),$$

where in the first sum we replaced every w_j by -1 and in the second we corrected the $j = 0$ term by $w_0 - (-1) = t + 1$. This equals $(-1)^d (1+t)$. Nothing in the argument used that t was an indeterminate, so it holds verbatim for a scalar c . Finally $\langle M'_d | \cdot \rangle = \langle s_{\mu_d}, \cdot \rangle$ by Lemma 8. $\square$

Corollary 12 (Q96: the answer is *no*). *For every $e \geq 2$ and every scalar $c \neq -1$ — and for the indeterminate t over $\mathbb{Q}(t)$ —*

$$\text{ord}_{\mathbb{Q}(t)[p_e]}(R_e(c)) = \text{ord}_{(e)}(R_e(c)) = \text{ord}(R_e(c)) = \infty.$$

At $c = -1$ all three are 0.

Proof. $(-1)^d (1+c) \neq 0$ for every d when $c \neq -1$, so $\text{ad}^d(R_e(c)) \neq 0$ for every $d \geq 1$; apply Corollary 7 and then Lemma 5. At $c = -1$, $R_e(-1) = M_{p_e}$ is a multiplication operator, of order 0 over any base. $\square$

So shrinking the base from Λ all the way down to $\mathbb{Q}(t)[p_e]$ — a single generator, the smallest base for which the question is not vacuous — does not make the order finite. The infinite order of $R_e(t)$ is not an artifact of measuring the deformation in all the p_m at once.

Remark 13 (the pin of the earlier proof was removable). The full-base result [2, Cor. 4.9] was proved by iterating $\text{ad}(M_{p_1})$, and $p_1 \notin \Lambda^{(e)}$ for $e \geq 2$; the entire mechanism of that proof is therefore unavailable here, which is why Q96 was open rather than a corollary. The outcome is that the pinned generator was *not* carrying the result: the same phenomenon is visible against M_{p_e} alone. Theorem 11 moreover recovers [2, Cor. 4.9] in one stroke and uniformly in the parameter — including the scalar $c = 0$, for which [2] needed a second theorem ([2, Thm. B]), because there the leading route weight degenerated. Here $c = 0$ gives $(-1)^d$, which is visibly nonzero.

Remark 14 (why the obvious channel is blind). It is worth recording the channel that does *not* work, since it is the first one to try. Take the target $((d+1)e)$, a single row, whose Maya set is $(\mathbb{Z}_{<0} \setminus \{-1\}) \cup \{(d+1)e-1\}$. The same flow argument shows the route is again unique — the bead at -1 walks to the right in $d+1$ hops — but now *every* window $(-1+ke, -1+(k+1)e)$ consists of sites ≥ 0 and is therefore empty, so all heights are 0, all weights are 1, and the matrix element is $\sum_j (-1)^{d-j} \binom{d}{j} = 0$ for every $d \geq 1$. This channel is constant in the direction it is meant to test: it cannot see t at all. Starting the walk at -2 instead of -1 is exactly what puts one bead, the one at -1 , inside the first window.

Remark 15 (independence from the LLT map). Every step above is a statement about legal e -moves on Maya sets, the height statistic $\#(M \cap (b, b+e))$, and a binomial identity. No LLT function, no q -deformation and no image under Lam’s map Φ appears anywhere. This matters because Φ sends the relevant operators to multiplication operators, which have order 0; an argument routed through Φ could not distinguish order 0 from order ∞ here. The present argument does not use it.

4 The two-parameter commutator

The $d=0$ instance of Corollary 12 is the statement $[R_e(t), R_e(-1)] \neq 0$: one operator, two values of the parameter. We record the general two-parameter commutator, which explains the shape of the obstruction and specialises correctly at $s=t$ and at $t=s=-1$. The derivation is the route enumeration of [3, Thm. 2.3] carried out with the two parameters kept separate: each route picks up each height-weight from whichever operator traverses that window.

Theorem 16 (matrix elements of $[R_e(t), R_f(s)]$). *Let $e, f \geq 1$, let M be a Maya set and $M' \neq M$. Then $\langle M' | [R_e(t), R_f(s)] | M \rangle = 0$ unless $|M \triangle M'| \in \{2, 4\}$, and:*

(1) One-bead sector. *If $M' = M \cdot (a, e+f)$ is a legal $(e+f)$ -move, write $N = \#(M \cap (a, a+e+f))$, $A = \#(M \cap (a, a+f))$, $B = \#(M \cap (a, a+e))$, and $m_e = \mathbf{1}[a+e \in M]$, $m_f = \mathbf{1}[a+f \in M]$. Then*

$$\langle M' | [R_e(t), R_f(s)] | M \rangle = s^A t^{N-1-A} (t - m_f(1+t)) - t^B s^{N-1-B} (s - m_e(1+s)).$$

(2) Two-bead sector. *If $|M \triangle M'| = 4$, write $M \setminus M' = \{b, c\}$. Call an ordered pair (b, c) (with $\{b, c\} = M \setminus M'$) a legal assignment if $b+e \in M' \setminus M$, $c+f \in M' \setminus M$ and $b, c, b+e, c+f$ are pairwise distinct. For such a pair put $P = \#(M \cap (b, b+e))$, $Q = \#(M \cap (c, c+f))$ and*

$$k = k_{b,c}^{e,f} = \mathbf{1}[b+e \in (c, c+f)] - \mathbf{1}[b \in (c, c+f)].$$

Then

$$\langle M' | [R_e(t), R_f(s)] | M \rangle = \sum_{(b,c) \text{ legal}} t^{P-k} s^Q (1 - (ts)^k).$$

There is exactly one legal assignment when $e \neq f$, and exactly two — (b, c) and (c, b) , whose indices k are negatives of one another — when $e = f$.

Proof. Both sectors are the enumeration of [3, Thm. 2.3]; only the bookkeeping of the parameters changes, so we give that.

(1) R_e first. As in [3, Thm. 2.3, Step 2], exactly two route types reach $M \cdot (a, e+f)$: the e -move from a followed by the f -move from $a+e$, which exists iff $m_e = 0$ and has heights B (for the e -move) and $N-B$ (for the f -move, using $m_e = 0$); and the e -move from $a+f$ followed by the

f -move from a , which exists iff $m_f = 1$ and has heights $N - 1 - A$ and A . Attaching t to e -moves and s to f -moves,

$$\langle M' | R_f(s) R_e(t) | M \rangle = (1 - m_e) t^B s^{N-B} + m_f t^{N-1-A} s^A.$$

Exchanging the roles of (e, t) and (f, s) — the target is unchanged —

$$\langle M' | R_e(t) R_f(s) | M \rangle = (1 - m_f) s^A t^{N-A} + m_e s^{N-1-B} t^B.$$

Subtract and group the two monomial types. The $s^A t^\bullet$ terms give $(1 - m_f) s^A t^{N-A} - m_f s^A t^{N-1-A} = s^A t^{N-1-A} (t - m_f(1 + t))$, using $m_f \in \{0, 1\}$; the $t^B s^\bullet$ terms give $m_e s^{N-1-B} t^B - (1 - m_e) t^B s^{N-B} = -t^B s^{N-1-B} (s - m_e(1 + s))$.

(2) Fix a legal assignment (b, c) : R_e moves b by e and R_f moves c by f . If the e -move goes first its height is P and the subsequent f -move sees a Maya set in which the count on $(c, c + f)$ has changed by exactly k , so its height is $Q + k$; the contribution to $R_f(s) R_e(t)$ is $t^P s^{Q+k}$. If the f -move goes first its height is Q and the subsequent e -move has height $P + (\beta - \alpha)$ where, by [3, Lem. 2.1], $\beta - \alpha = -k$ because $b \neq c$ and $b + e \neq c + f$; the contribution to $R_e(t) R_f(s)$ is $s^Q t^{P-k}$. The net contribution to the commutator is $s^Q t^{P-k} - t^P s^{Q+k} = t^{P-k} s^Q (1 - (ts)^k)$. Summing over legal assignments gives the formula.

For the count of legal assignments: the bead moved by R_e is displaced by e and the one moved by R_f by f . If $e \neq f$ this determines which is which, since the alternative would force $c - b = f - e$ and $c - b = e - f$ simultaneously. If $e = f$ both assignments are available, and interchanging b and c replaces $k_{b,c}^{e,e}$ by $k_{c,b}^{e,e} = -k$. $\square$

Corollary 17 (the specialisations).

- (i) At $s = t$ the formulas reduce to [3, Thm. 2.3]: sector (1) becomes $(m_e - m_f)(1 + t)t^{N-1}$ and sector (2) becomes $t^{P+Q}(t^{-k} - t^k)$ summed over legal assignments, which cancels in pairs when $e = f$, consistently with $[R_e(t), R_e(t)] = 0$.
- (ii) At $t = s = -1$ both sectors vanish identically, consistently with the fact that all M_{p_k} commute.
- (iii) The two-bead sector vanishes identically if and only if $ts = 1$.
- (iv) $[R_e(t), R_e(s)]$ has a nonzero two-bead part whenever $t \neq s$ and $ts \neq 1$.

Proof. (i) In sector (1) put $s = t$: the first term is $t^{N-1}(t - m_f(1 + t))$ and the second is $t^{N-1}(t - m_e(1 + t))$, whose difference is $t^{N-1}(1 + t)(m_e - m_f)$. In sector (2), $t^{P-k}t^Q(1 - t^{2k}) = t^{P+Q}(t^{-k} - t^k)$; at $e = f$ the two legal assignments contribute $t^{P+Q}(t^{-k} - t^k)$ and $t^{P+Q}(t^k - t^{-k})$.

(ii) At $t = -1$ we have $t - m_f(1 + t) = -1$ and likewise $s - m_e(1 + s) = -1$, so sector (1) becomes $(-1)^A(-1)^{N-1-A}(-1) - (-1)^B(-1)^{N-1-B}(-1) = 0$; and $ts = 1$ kills sector (2) by (iii).

(iii) Each summand carries the factor $1 - (ts)^k$, which vanishes when $ts = 1$. Conversely $k = \pm 1$ does occur (for instance $e = 1$, $f = 2$ and a configuration with $b \in (c, c + f)$), and there $1 - (ts)^{\pm 1} \neq 0$ unless $ts = 1$.

(iv) Immediate from (iii) and the two-assignment description: at $e = f$ the two summands are $t^{P-k} s^Q (1 - (ts)^k)$ and $t^{Q+k} s^P (1 - (ts)^{-k})$, whose sum is $t^{P-k} s^Q - t^P s^{Q+k} + t^{Q+k} s^P - t^Q s^{P-k}$; this is not identically zero. $\square$

Remark 18 (a correction to a case split). Part (iv) corrects a reading of [3, Thm. 2.3(2)], whose clause “if $e = f$ then the matrix element is 0” is a *one-parameter* statement: it holds because the two legal assignments cancel when $s = t$, not because the sector is empty. Once $t \neq s$ the

cancellation fails. Concretely, for $e = f = 3$, $\lambda = \emptyset$, $b = 6$, $c = 8$ in a 9-bead window one finds $P = 2$, $Q = 0$, $k = 1$ and matrix element $s^2t - st^2 - s + t = (t - s)(1 - st)$, which is nonzero off the two loci $s = t$ and $st = 1$. This was found by the numerical sweep of §5, which reported exactly 61 disagreements against a first version of the formula that had imported the $e = f$ clause verbatim; all 61 were $e = f$ two-bead entries.

Remark 19 (why the one-bead sector no longer factors). In operator form the one-bead sector of $[R_e(t), R_f(s)]$ is

$$\sum_{a \in \mathbb{Z}} \psi_{a+e+f} \psi_a^* \left[(-s)^{N(a, a+f)} (-t)^{N(a+f, a+e+f)} - (-t)^{N(a, a+e)} (-s)^{N(a+e, a+e+f)} \right],$$

where $N_{(x,y)} = \sum_{x < j < y} n_j$. Each term dresses the interval $(a, a + e + f)$ in *two colours*: the sub-interval traversed by R_e carries t , the one traversed by R_f carries s — and the two orderings split the interval at different points ($a + f$ versus $a + e$). When $s = t$ the colours merge and both terms collapse to $(-t)^{N(a, a+e+f)}$ times a local factor, which is how the one-parameter answer acquires its clean form $-\frac{1+t}{t} \Phi_{e,f}$. When $s \neq t$ no such collapse occurs. This is the structural reason a change of parameter cannot be gauged away, and it is the mechanism behind Theorem 11: in the witness channel the first window (the one containing the bead -1) is traversed by $R_e(t)$ in exactly one of the $d + 1$ routes, and by M_{p_e} in all the others.

5 Verification

All computations were run with two code-disjoint engines and, for Proposition 3, a third.

1. **Engine A** (abacus) versus **engine B** (brute-force enumeration of border strips on Young diagrams, with explicit 2×2 and connectivity tests): $R_g(t)s_\lambda$ agrees on 60/60 pairs, $1 \leq g \leq 5$, λ ranging over 12 partitions with $|\lambda| \leq 8$, with t an indeterminate.
2. **Premise** $M_{p_g} = R_g(-1)$ (Proposition 3), re-verified against **engine C**: s_λ built by Jacobi-Trudi as a polynomial in $p_1, p_2, \dots$, multiplied by the *symbol* p_g , and re-expanded in the Schur basis by solving a linear system over the p -monomial basis. Agreement on 29/29 cases, $1 \leq g \leq 4$ and $|\lambda| + g \leq 7$. Engine C shares no code with A or B.
3. **Theorem 11**: the predicted value $(-1)^d(1 + t)$ was written down from Lemmas 9–10 *before* being computed, and then confirmed on 13/13 cases: (e, d) with $2 \leq e \leq 5$, $1 \leq d \leq 5$ and $(d + 1)e \leq 15$.
4. **Theorem 11, independent path**: the same 5 cases ($e = 2, d \leq 3$ and $e = 3, d \leq 2$) recomputed with engines B and C only — border strips on Young diagrams for $R_e(t)$, literal multiplication by p_e in $\mathbb{Q}(t)[p_1, p_2, \dots]$ for M_{p_e} — with no abacus anywhere in the path. 5/5.
5. **Scalar specialisations**: $c \in \{0, 1, 2, -\frac{1}{2}\}$ give $(-1)^d(1 + c)$ on all 6 available (e, d) ; $c = -1$ gives 0 on all 6.
6. **Theorem 16**: predicted versus computed matrix elements of $[R_e(t), R_f(s)]$, agreement on **1829/1829** entries. The denominator factors as 1232 one-bead entries + 597 two-bead entries, of which 61 have $e = f$; those 61 are exactly the entries that a version of the formula importing the one-parameter $e = f$ clause got wrong (Remark 18). Quantifier: $1 \leq e, f \leq 4$; all 30 partitions with $|\lambda| \leq 6$; every nonzero target of $[R_e(t), R_f(s)]s_\lambda$.

7. **Corollary 17:** specialising $s \rightarrow t$ reproduces an independently computed one-parameter commutator on 304/304 source states ($1 \leq e, f \leq 4, |\lambda| \leq 5$); $t = s = -1$ annihilates all 304; $s = 1/t$ annihilates all 379 two-bead entries.

Negative controls

Both controls move, i.e. each returns a different answer from the one the theorem returns.

- $e = 1$. Here $\Lambda^{(1)} = \Lambda$ and the theorem must *not* apply. It does not: a border strip of size 1 is a single cell, of height 0, so $R_1(t) = R_1(-1) = M_{p_1}$ is independent of t and has order 0 over every base. Checked: every coefficient of $R_1(t)s_\lambda$ equals 1 for all $|\lambda| \leq 7$; $[R_1(t), M_{p_m}] = 0$ for $1 \leq m \leq 4$ and all $|\lambda| \leq 5$; and the same machinery that produces $(-1)^d(1+t)$ for $e \geq 2$ returns $\text{ad}^d = 0$ at $e = 1$. The proof localises the failure exactly: Lemma 10 needs $e \geq 2$ for -1 to lie in the first window.
- $t = -1$. $\text{ad}(M_{p_e})^d(R_e(-1)) = 0$ for $e \in \{2, 3\}$, $1 \leq d \leq 3$ and all $|\lambda| \leq 4$, against $(-1)^d(1+t) \neq 0$ for generic t . The machinery therefore distinguishes the anchor from its neighbourhood, and is not constant in the direction it is meant to test.

Remark 20 (an instrument failure, recorded). The first run of verification item 4 disagreed with item 3, returning -2 where the theorem predicts $-t - 1$. The disagreement was in the instrument: the routine extracting p -monomial coefficients used `as_coefficients_dict`, which returns only the *numeric* part of a coefficient, silently discarding the symbolic factor t — so it was reporting the value at $t = 1$, and indeed $-t - 1$ at $t = 1$ is -2 . After the extractor was rewritten to make the p_k the polynomial generators and keep the full coefficient, all 5 cases agreed. Verification items 1 and 2 were unaffected, since they involve no symbolic coefficients ($t = -1$ there).

6 What is settled and what is not

Settled. Q96 is answered, in the negative and in the strong form: the order is infinite already over $\mathbb{Q}(t)[p_e]$ (Corollary 12). The reduction to algebra generators is proved, not imported (Lemma 6). The two-parameter commutator is in closed form (Theorem 16) with its specialisations checked (Corollary 17).

Consequence for the programme. The reading of Q96 under which a “yes” would have given Q91 a conditional normal form over $\Lambda^{(e)}$ is closed off: the obstruction is not removable by restricting the base. Whatever normal form $R_e(t)$ has, it is not one with finitely many $\partial_{p_{me}}$.

Not settled, and not attempted here.

1. $\text{ord}_{(e)}(R_g(t))$ for g not a multiple of e , and more generally the whole two-index table. The flow argument of Lemma 9 uses that every move has the *same* size e ; with two step sizes the multiset of moves is no longer forced by divergence alone and the route count is genuinely harder.
2. The locus $ts = 1$ of Corollary 17(iii). It is a second distinguished locus for the two-parameter family, alongside $s = t$ and the anchor $t = s = -1$ (which lies on both). We have proved only that the two-bead sector dies there; what the surviving one-bead operator is, and whether the locus means anything, is not examined.
3. Nothing in this note is a claim about the literature. No source was read during this session beyond the author’s own earlier files, and no priority or novelty assertion is made about any statement above.

References

- [1] Clio, *The multiplication defect of the ribbon operator* (Q75), `proofs/2026-09-03-Q75-multiplication-defect.tex`, 3 September 2026.
- [2] Clio, *$\text{ord}(N_e) = \infty$, and the order filtration is discontinuous at $t = -1$* (Q84), `proofs/2026-09-06-Q84-order-of-N-e.tex`, 6 September 2026.
- [3] Clio, *The cross-rank commutator $[R_e(t), R_f(t)]$ in closed form* (Q92), `proofs/2026-09-07-Q92-cross-rank-commutator.tex`, 7 September 2026.